MIL-T-36213A
18 June 1966
SUPERSEDING
MIL-T-36213(DM)
25 May 1964

MILITARY SPECIFICATION

TANK, INSERT, X-RAY FILM PROCESSING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers two types of corrosion-resisting steel insert tanks. One type shall be mounted on legs and shall be suitable for use with two and three compartment x-ray film processing units. The other type shall be lightweight and suitable for field use with a single compartment x-ray film processing master tank.

1.2 Classification.

1.2.1 Types. The insert tanks shall be of the following types:

Type I. With legs.
Type II. Without legs.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6525

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:13Z
Check the source to verify that this is the current version before use.

STANDARDS

Federal

FED-STD-66	Steel, Chemical Composition and Hardenability.
FED-STD-151	Metals, Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWINGS

Defense Personnel Support Center

20495	Tank Insert, X-Ray Film Processing, Corrosion-Resisting Steel.
20678	Tank Insert, X-Ray Film Processing, Corrosion-Resisting Steel, 5 Gal.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), prior to the start of production of the contract quantity, one preproduction sample of the insert tank shall be assembled, completely finished, and tested by the contractor. The sample shall be submitted, together with the contractor's inspection and test reports, to the contracting officer for examination and testing to determine compliance with the requirements of this specification. Written approval of the materials, construction, and performance of the sample shall be obtained from the contracting officer, or his authorized representative, prior to the manufacture and assembly of the remaining quantity to be manufactured. Further production of the item by the contractor, prior to this approval, shall be at the contractor's risk. Accepted preproduction sample shall become the property of the procuring activity and shall be included in the quantity called for in the contract or order. In the event that the preproduction sample is not approved, the contractor shall submit a new sample for approval, or modify the existent sample so as to remove the deficiencies cited.

3.2 Material. The insert tanks shall be fabricated from AISI 316L corrosion-resisting steel conforming to the chemical requirements of FED-STD-66 when tested as specified in 4.4.1.

3.2.1 Type I. Type I tanks shall be 0.050 inch (No. 18, United States Standard (U.S.S.) gage).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:13Z
Check the source to verify that this is the current version before use.

3.2.2 Type II. Type II tanks shall be 0.025 inch (No. 24 U.S.S. gage).

3.3 Construction. The insert tanks shall be of rigid, durable, leakproof construction. The outside surfaces of the tank shall not distort more than 0.015 inch per linear inch of the surface when hydrostatically tested for rigidity as specified in 4.4.2. When tested for capacity, as specified in 4.4.3, the volume of water required to fill the insert tank to a level of one inch below the top shall be between 100 percent and 103 percent of rated capacity. Finish shall be Commercial Standard No. 2B. Instructions, which explain the importance of and describe the method for keeping the tanks clean, shall be placed in each tank.

3.3.1 Type I. Design and dimensions of Type I tanks shall be as shown on DPSC Drawing 20495, revised 8 June 1966.

3.3.2 Type II. Design and dimensions of Type II tanks shall be as shown on DPSC Drawing 20678, revised 8 June 1966.

3.4 Identification marking. Each tank shall be permanently and legibly marked with the name or registered trademark of the manufacturer, and with the rated capacity of the tank. In addition, each tank shall be stamped with the letters"U.S." 1 inch or more in height. Marking secured by adhesives shall not be used.

3.5 Workmanship. Workmanship shall be first class throughout. All interior surfaces shall be smooth and free of pits, protruding ridges, and other defects which detract from their appearance or may impair their serviceability.

3.5.1 Welding. Except as otherwise indicated on the drawing, welding of corrosion-resisting steel shall be by the inert gas (heliarc) method. Complete penetration of corner welds shall be accomplished to provide a sound weld and to eliminate the crevice between the two welded sheets. Where welding rod is used, it shall be 316L, and the resultant weld shall be ground and polished smooth to remove all pits, discontinuities, flux, scale, and laps. Tanks shall pass the test for leakage specified in 4.4.4. Spot welds shall be smoothly dressed to blend with adjacent surfaces.

3.5.2 Passivation of welds. After fabrication, as a final treatment, tanks shall be passivated or pickled by complete immersion into a bath of nitric acid or a mixture of nitric and hydrofluoric acids of proper concentration and temperature. Slight etching shall be permitted on the number 2B finish as a result of the passivation or pickling process to remove contamination and discoloration. Following the passivation or pickling treatment, tanks shall be thoroughly washed with water until all traces of the acid solution have been removed, and shall be dried. There shall be no evidence of corrosion when tanks are subjected to the corrosion resistance test specified in 4.4.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:13Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Military Standard MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with Military Standard MIL-STD-105, inspection level S4, except that the acceptance number shall be zero for all sample sizes and a minimum of three samples shall be taken regardless of lot size.

4.2.3 Raw materials. Four ounces of each lot of each metal used in the construction of the tanks shall be selected for chemical analysis.

4.3 Examination. The tanks shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Chemical composition. Tests for chemical analysis shall be conducted in accordance with method 111.1 or 112.1 of Federal Test Method Standard 151.

4.4.2 Hydrostatic test for rigidity. The tank under test shall be placed on a flat surface. The camber of each outside surface shall be measured by the use of a straight edge applied horizontally, vertically, and diagonally across the center of each side. Fill the tank with water. Remeasure the camber. Check for leakage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:13Z
Check the source to verify that this is the current version before use.

4.4.3 Capacity test. Measurement to determine capacity of the insert tank shall be made as follows, in the order listed:

(a) Place the insert tank into a larger tank or within a master tank.
(b) From a premeasured volume of water (at approximately 70° F.) equal to the rated capacity of the insert tank, pour approximately 80% into the insert tank.
(c) Fill the larger or master tank with water to a level approximately one inch below the top of the insert tank.
(d) Add the balance of the premeasured volume of water (in (b) above) to the insert tank.
(e) Measure the distance from the top to the water level in the insert tank.
(f) Pour an additional volume of water into the insert tank equal to three percent of its capacity.
(g) Repeat measurement made in (e) above.
(h) Analyze measurements obtained in (e) and (g) above to determine whether the one inch water level below the top of the tank occurs between 100 percent and 103 percent of rated capacity (see 3.3).

4.4.4 Tank leakage. All welds on the tank shall be coated on the outside with a fluorescent penetrant suspended in oil. The welds shall be viewed from the inside of the tank under ultraviolet light in a darkroom for indications of porosity and breaks in the welds.

4.4.5 Corrosion resistance. A 3-5 percent salt solution maintained at 140° F. shall be allowed to stand in the tank for 48 hours. The tank shall be observed for evidence of corrosion.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One tank, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I. Packaging Quantities

Size	Unit package	Exterior container
5 gal	1 unit	2 units
10 gal	1 unit	1 unit
15 gal	1 unit	1 unit
20 gal	1 unit	1 unit

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:13Z
Check the source to verify that this is the current version before use.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to the contents. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C.

5.1.4.1 Unit shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:13Z
Check the source to verify that this is the current version before use.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Type I insert tanks shall be suitable for use with FSN 6525-616-7651 PROCESSING UNIT, X-RAY FILM, 2 Compartment, with Rear Panel and FSN 6525-616-7300 PROCESSING UNIT, X-RAY FILM, 3 Compartment, with Rear Panel. Type II insert tanks shall be suitable for use with FSN 6525-616-7400 TANK, MASTER, X-RAY FILM PROCESSING, Refrigerated, Single Compartment, 110 volt, 60 cycle, AC.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Type and capacity of insert tank required (1.2).
(c) When preproduction sample is not required (3.1).
(d) Where preproduction sample shall be sent (3.1).
(e) Selection of applicable levels of packing (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

6.4 This specification covers the following items:

Stock No.		Item Identification.
6525-604-2150	(Type I)	TANK, INSERT, X-RAY FILM PROCESSING, Corrosion-Resisting Steel, 5 gal
6525-604-2155	(Type I)	TANK, INSERT, X-RAY FILM PROCESSING, Corrosion-Resisting Steel, 10 gal
6525-604-2160	(Type I)	TANK, INSERT, X-RAY FILM PROCESSING, Corrosion-Resisting Steel, 15 gal
6525-604-2165	(Type I)	TANK, INSERT, X-RAY FILM PROCESSING, Corrosion-Resisting Steel, 20 gal
6525-616-7679	(Type II)	TANK, INSERT, X-RAY FILM PROCESSING, Corrosion-Resisting Steel, 5 gal

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:13Z
Check the source to verify that this is the current version before use.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

CATEGORY N

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:13Z
Check the source to verify that this is the current version before use.

CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

FOLD UNDER

POSTAGE FEES PAID
DEFENSE SUPPLY AGENCY

DEFENSE SUPPLY AGENCY
DEFENSE PERSONNEL SUPPORT CENTER
2800 SOUTH 20th ST
PHILADELPHIA, PA. 19101

OFFICIAL BUSINESS

DEFENSE SUPPLY AGENCY

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER

ATTN: DIRECTORATE MEDICAL MATERIEL

2800 SOUTH 20TH STREET

PHILADELPHIA, PA. 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:13Z
Check the source to verify that this is the current version before use.

s No. 11A 0101 80; 3000

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-T-36213A

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

0

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES. ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1426

SND PPSO 12302

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:13Z
Check the source to verify that this is the current version before use.